

The quadratic formula

WORKSHEET



Interactive version

Name _____

Class _____

Date _____

WORKED EXAMPLES

Three worked examples

Example 2 · The second garden: $x^2 + 3x - 30 = 0$

COEFFICIENTS

$$a = 1, b = 3, c = -30.$$

DISCRIMINANT

$$D = 3^2 - 4 \cdot 1 \cdot (-30) = 9 + 120 = 129 > 0.$$

FORMULA

$$x_{1,2} = \frac{-3 \pm \sqrt{129}}{2}, \text{ so } x_1 \approx 4.18 \text{ and } x_2 \approx -7.18.$$

ANSWER · YOUR TURN

Example 3 · One root: $4x^2 - 4x + 1 = 0$

COEFFICIENTS

$$a = 4, b = -4, c = 1.$$

DISCRIMINANT

$$D = (-4)^2 - 4 \cdot 4 \cdot 1 = 16 - 16 = 0, \text{ so there is one root.}$$

ROOT · YOUR TURN

CHECK · YOUR TURN

CHECK

Check yourself

Level 1 · threshold

1.

How many real roots does $3x^2 - 5x + 4 = 0$ have?

- ☐ A one
 ☐ B none
 ☐ C can't tell without solving
 ☐ D two

2.

Solve $x^2 - 5x - 14 = 0$. Give both roots, separated by a semicolon.

Answer:

3.

Put the steps for solving $2x^2 + x - 6 = 0$ in the right order.

☐

Read off $a = 2$, $b = 1$, $c = -6$.

☐

Compute $D = 1^2 - 4 \cdot 2 \cdot (-6) = 49$.

☐

$x_{1,2} = \frac{-1 \pm 7}{4}$.

☐

$D > 0$, so there are two roots.

☐

$x_1 = \frac{3}{2}$, $x_2 = -2$.

4.

For which value of c does $x^2 + 6x + c = 0$ have exactly one root?

Answer:

5.

Which equations have two different real roots? Choose all that apply.

☐ A $x^2 - 4x + 4 = 0$

☐ B $x^2 + 4 = 0$

☐ C $-x^2 + 3x + 1 = 0$

☐ D $x^2 - 4 = 0$

EXAM TASK

Two consecutive integers

Level 3 · basic

PUPP style · 4 points

The sum of the squares of two consecutive positive integers is 181. Find the integers.

